UNIVERSITY OF ALBERTA

Engineering
Co-op Program

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ACADEMIC & CO-OP STATUS

Electrical Engineering, BSc Co-op, University of Alberta — Class of 2029

Completed Academic Terms	5 of 8
Completed Co-op Work Terms	0 of 5
Availability Starting January 2027	4 or 8 months

First Class Standing (3.5+ GPA) · EECERT Lab Certification · Class 5 licence · Willing to relocate

EXPERIENCE

Electrical Apprentice *Powerworks, Edmonton, AB* — Summer 2026

- Programmed the cause-and-effect logic on a Mircom FX-3500 addressable fire alarm panel for a $9.6M three-storey commercial building, mapping pull stations and smoke and heat detectors to the elevator recall relays that send cars to a safe floor and hold them on alarm
- Found an intermittent ground fault on a live fire alarm circuit by splitting the loop at the panel's terminal blocks and ohming out each half until the bad segment showed up, then repaired it without pulling new cable
- Walked the finished building with the consulting engineer for the fire alarm verification inspection (CAN/ULC-S537) and cleared the deficiencies before sign-off
- Read single-line diagrams, panel schedules and device layouts across three floors, and set the install order and circuit routing for the crew from those drawings

Residential Solar Sales (door to door, commission) *Orizon Energy* — Part-time, weekends

- Pitched rooftop solar door to door, talking with about 25 homeowners a day; learned to find out quickly what each client cared about and explain the system in plain terms

ENGINEERING TEAM

Electrical Subsystem Engineer *ARVP, University of Alberta* — Sept. 2025 – Apr. 2026

- Designed and had fabricated the team's electrical onboarding board in Altium: a Teensy 4.0 carrier with a CAN transceiver and 120 Ω termination, an I²C temperature sensor, a Hall-effect input with LED indicator, and an RC-filtered op-amp buffer on the analog input
- Documented every interface on the comms board with a teammate (CAN, Jetson Orin header, payload ports); it became the requirements for the stripped-down next revision

Avionics Member (FPGA) *STARR Rocketry, University of Alberta* — Sept. 2026 – Present

- Developing FPGA logic for the rocket's avionics, working across sub-teams with the liquid propulsion and recovery groups

GeoWall Design Competition *1st Place and Best Presentation* — Feb. 2025

- Beat fourth-year civil engineering teams: planned the structure two weeks out with drawings, material choices and kPa load predictions, and our wall took the highest hydraulic press load of the competition

PROJECTS

Adaptive LED Lamp *ESP32, C++, Python, BH1750, PWM* — Aug 2026 – Present

- Built and characterized an ESP32 lamp that PWM-drives a 12 V LED strip through a logic-level MOSFET; measured light output is linear in duty cycle (R^2 = 0.9999 from 10–100 %) with no dead zone at low brightness
- Added a closed-loop constant-light mode: the lamp measures its own gain with the light sensor, then an integral controller holds the target brightness, settling in under 5 s with under 0.25 lux overshoot (stability checked in simulation first)

STM32F411 IMU Board *Altium, STM32, I²C, SWD* — Dec 2025

- Designed a 4-layer STM32F411 board on my own to learn the chip without a dev board: USB power through a ferrite bead and 3.3 V LDO, 24 MHz crystal, reset and boot straps, SWD debug, and an MPU-6050 IMU on I²C (design only, not fabricated)

SKILLS

Hardware: Altium Designer (2- and 4-layer PCB), LTspice, Schematic review, Soldering (through-hole, SMD), Oscilloscope, Multimeter
Firmware: C/C++, ESP32 (Arduino core), STM32, Teensy, PWM, ADC, I²C, SPI, UART, CAN bus
Software: Python (NumPy, Matplotlib, pySerial), MATLAB, VHDL (Zybo Z7 FPGA), Git
Field: Canadian Electrical Code, Single-line diagrams, 347 V and 120/208 V, Overcurrent protection, Fire alarm programming (Mircom), Revit lighting (learning)